

CMS $137.6 \text{ fb}^{-1} (13 \text{ TeV})$

2HDM+S Type-IV

 $m_H = 125 \text{ GeV}$

95% CL upper limits

--- Expected exclusion

— Observed exclusion

--- Expected exclusion

— Observed exclusion

--- Expected exclusion

— Observed exclusion

--- Expected exclusion

— Observed exclusion

 $\frac{\sigma_H B(H \rightarrow aa)}{\sigma_{SM}}$ 10^{-1} 10^{-2} 10^{-3} $m_a \text{ (GeV)}$ $\tan \beta$ 